CRYPTOGRAPHY | FINAL PROJECT REPORT

Bitcoin Real-Time Cryptographic Dashboard

A live monitor of Proof-of-Work, block headers, network difficulty and an AI forecast of the next difficulty adjustment.

AUTHOR	María Muñoz Nadales
COURSE	Cryptography
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1 Project Overview

The *Bitcoin Real-Time Cryptographic Dashboard* is an interactive Streamlit application that visualises live data from the Bitcoin network and explains, step by step, the cryptographic primitives that make the chain trustworthy. It connects to public Bitcoin APIs (`blockchain.info` and `Blockstream Esplora`) and is organised in four core modules (**M1**–**M4**) plus four advanced ones (**M5**–**M8**). The dashboard is centred on three questions that map directly to the topics of the course: **(i)** how is each block protected by hashing, **(ii)** how does the network adapt difficulty so that a new block is produced roughly every ten minutes, and **(iii)** how can a simple AI model anticipate the next difficulty adjustment.

Module	Displayed evidence	Cryptographic meaning
M1	Difficulty, block time, hash-rate estimate	Measures the amount of work securing the current chain tip.
M2	Header fields, target, nonce, hash validation	Verifies Proof of Work locally from the 80-byte block header.
M3	Historical difficulty and recent trend	Shows how Bitcoin retargets mining cost over time.
M4/M7	Difficulty forecast and backtest MAE	Evaluates a predictor for the next difficulty point.
M5/M6	Merkle proof and attack probability	Connects transaction integrity with confirmation security.

2 Cryptographic Metrics Displayed

The dashboard displays a set of metrics that together describe the security state of the Bitcoin chain at a given moment. Each one is computed in real time and explained inside the UI.

2.1 Block hash and double-SHA-256

For every new block, module M2 fetches the raw 80-byte block header and recomputes the hash locally as $H = \mathrm{SHA256}(\mathrm{SHA256}(\mathrm{header}))$, reversing the output to display order. This is the same operation Bitcoin miners repeat billions of times per second. The dashboard parses the six header fields (`version`, `previous_block_hash`, `merkle_root`, `timestamp`, `bits`, `nonce`) and verifies that the recomputed hash matches the value reported by the API, proving header integrity client-side.

2.2 Target, leading zero bits and Proof of Work

The compact `bits` field is unpacked into the full 256-bit target T through $T = c \cdot 2^{8(e-3)}$, where e is the exponent and c the coefficient. A block is valid when the hash interpreted as an integer satisfies $H \leq T$. The UI shows the number of *leading zero bits* of H, which is the most intuitive proxy for the work spent: each extra leading zero requires, on average, twice as many hash attempts. This is the practical realisation of Adam Back's *hashcash*-style proof of work referenced in the Bitcoin whitepaper [1].

2.3 Difficulty and estimated hash rate

Difficulty D is reported per block. Combined with the average inter-block time $\bar{t}$ measured over the last ten blocks, module M1 estimates the network hash rate as

$$R = \frac{D \cdot 2^{32}}{\bar{t}}$$

This formula comes from the fact that, on average, $D \cdot 2^{32}$ hashes must be tried to find a valid block. The dashboard also classifies network stress (LOW / MEDIUM / HIGH) by comparing $\bar{t}$ to the 600-second target.

2.4 Difficulty history and re-targeting

Module M3 requests the public `blockchain.info/charts/difficulty` series and displays the last 180 sampled difficulty points used by the dashboard. Bitcoin re-targets every 2016 blocks (∼two weeks); the chart reveals these step-like adjustments and the medium-term trend of network commitment. KPIs include latest, previous, minimum and maximum difficulty over the selected window.

2.5 Merkle proof and 51% attack economics (advanced)

Module M5 reconstructs the Merkle tree of the latest block from its transaction IDs and walks the path from any chosen transaction up to the `merkle_root`, validating it with double-SHA-256. Module M6 implements Nakamoto's catch-up probability for an attacker controlling a fraction q of the hash rate after z confirmations [1, §11]:

$$P(z,q) \;=\; 1 - \sum_{k=0}^{z} \frac{\lambda^k e^{-\lambda}}{k!} \left(1 - \left(\tfrac{q}{p}\right)^{z-k}\right), \quad \lambda = z \cdot \tfrac{q}{p}.$$

This curve, together with an electricity-cost model, lets the user explore how many confirmations are economically safe.

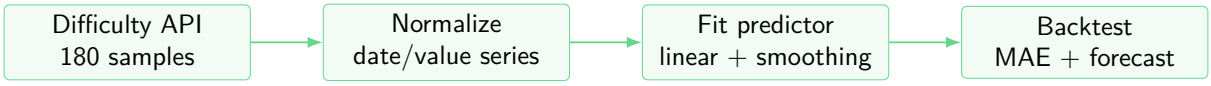
3 AI Component

3.1 Model and rationale

The chosen AI approach is a **Predictor** for the next Bitcoin difficulty value, implemented in module M4. The primary model is an **ordinary least-squares linear regression** fitted on the same last 180 sampled difficulty points used by the dashboard, returning slope β and intercept α so that $\hat{D}_{n+1} = \alpha + \beta\, n$. A second model (M7) uses **simple exponential smoothing** with weight α, computing $S_t = \alpha\, D_t + (1-\alpha)\, S_{t-1}$ and using the last S as the next-step forecast.

Linear regression was selected as the main predictor for three practical reasons:

- Bitcoin difficulty changes in visible retargeting steps but often has a clear medium-term direction; a first-order trend is a transparent approximation of that direction.
- It is fully transparent: slope and intercept can be inspected by students of the course, which matches the pedagogical goal of the dashboard.
- It runs in milliseconds and has no external dependency, so it can be re-trained on every refresh without slowing the UI.



3.2 Evaluation results

Both models are evaluated with a rolling **one-step-ahead backtest**: at each index $t \geq 8$, the model is retrained on $D_{0..t-1}$ and the absolute error $|\hat{D}_t - D_t|$ is recorded. The reported metric is the *Mean Absolute Error* (MAE) over the rolling window.

Model	**Rolling MAE**	**Last-step error**
Linear regression (180 samples)	3.66×10^{12}	0.56 % relative
Exponential smoothing (α=0.35)	1.18×10^{12}	0.44 % relative

These values were computed from the live dataset on 7 May 2026. In that run, exponential smoothing achieved the lower rolling MAE because it follows recent retargeting steps more closely. Linear regression remains the main M4 model because it makes the trend direction explicit and

is easier to explain in a cryptography dashboard; M7 is included to show the trade-off and avoid relying on a single untested predictor. The dashboard renders a *confidence band* proportional to the predicted percentage change and labels the forecast as High / Medium / Low confidence based on the magnitude of $\hat{D}_{n+1} - D_n$.

4 Conclusions

The dashboard turns abstract concepts (hash, target, Merkle tree, attack probability) into measurable, live quantities. The cryptographic layer is verified locally rather than trusted from the API: every header is re-hashed, every Merkle root is rebuilt from raw transaction IDs, and every target is decoded from `bits`. The AI layer is intentionally simple to remain interpretable, while the side-by-side comparison in M7 shows that model choice matters and must be evaluated rather than assumed. Future work would promote the lower-error model when it consistently wins and test an ARIMA/Prophet variant that explicitly captures the 2016-block re-targeting period.

References

[1] S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008.
https://bitcoin.org/bitcoin.pdf